



DIRECT PREFERENCE OPTIMIZATION: ALIGNING GENERATIVE AI WITH HUMANS, FROM TEXT TO IMAGES



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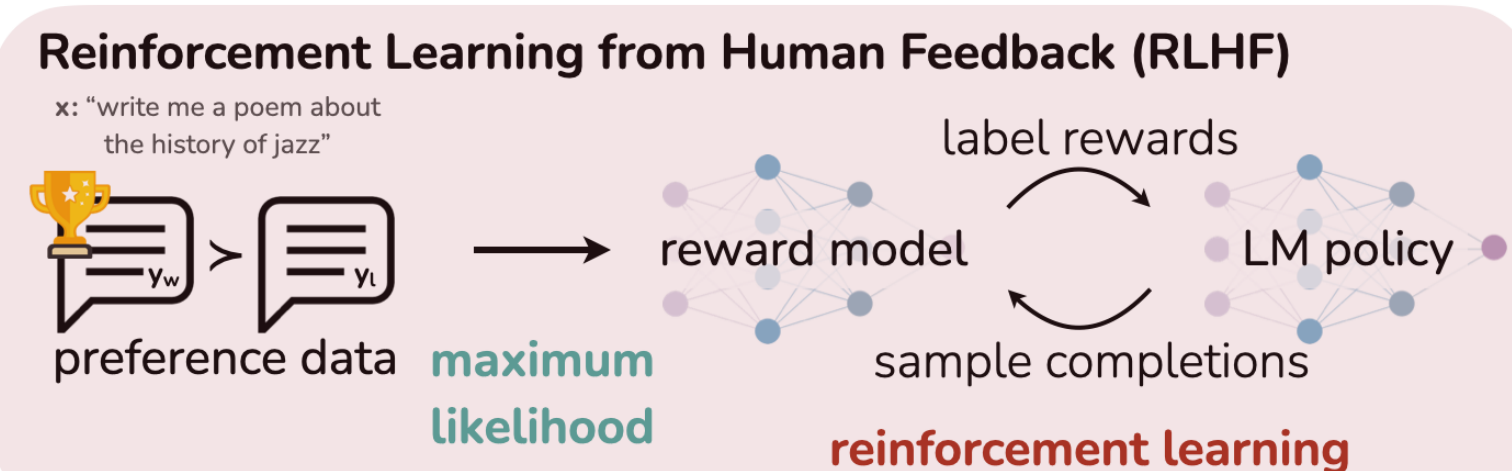
INTRODUCTION

We may not want our generative model imitating some of the skills in it's training data.

How do we efficiently fine-tune these models using preference data?

Given prompt x , humans prefer output y^w over y^l .
($y^w \succ y^l | x$)

RLHF



The Bradley-Terry (BT) model:

$$p_{BT}(y^w \succ y^l | x) = \sigma(r(x, y^w) - r(x, y^l))$$

where σ is the sigmoid function.

$$L_{BT}(\phi) = -\mathbb{E}_{x, y^w, y^l} [\log \sigma(r_\phi(x, y^w) - r_\phi(x, y^l))]$$

RL objective:

$$\max_{p_\theta} \mathbb{E}_{x \sim \mathcal{D}, y \sim \pi_\theta(y|x)} [r(x, y)] - \beta \mathbb{D}_{KL}[\pi_\theta(y|x) \parallel \pi_{ref}(y|x)]$$

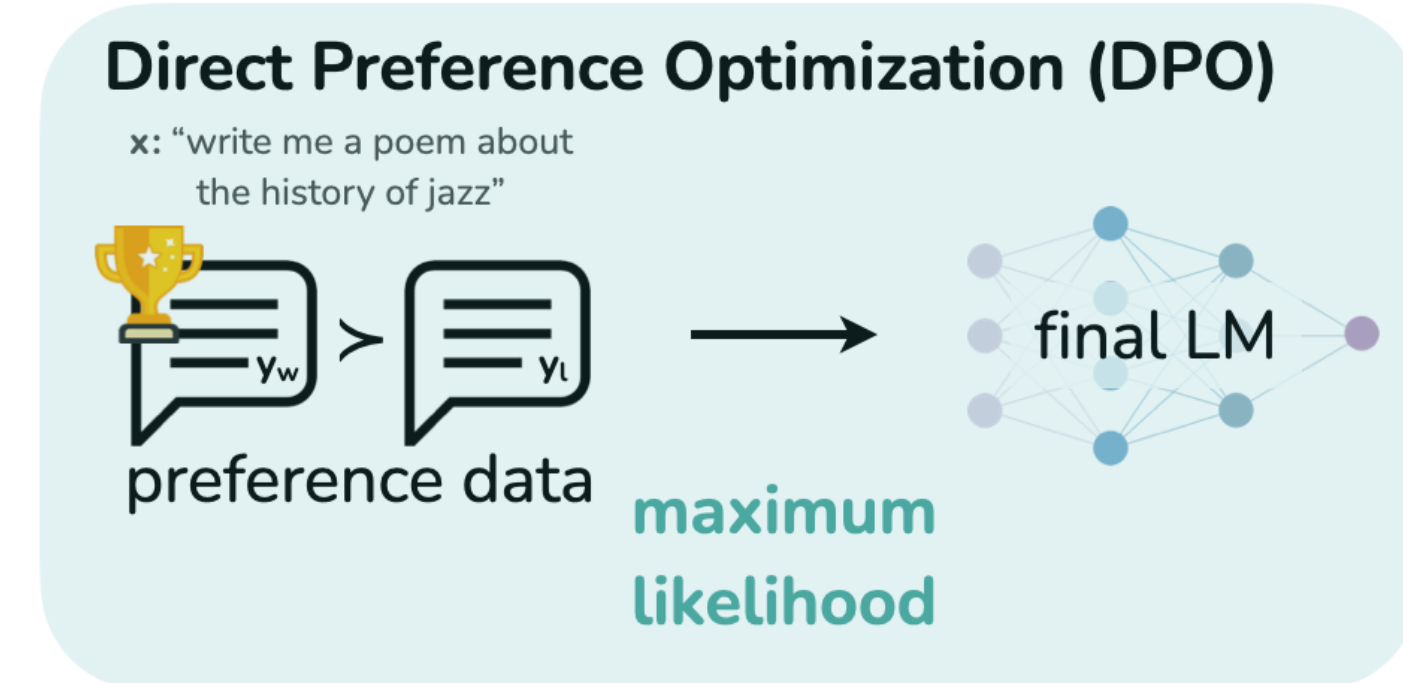
β controls regularization.

Training Instability, Complex Pipeline, Resource Intensive

REFERENCES

- [1] Rafael Rafailov, Archit Sharma, Eric Mitchell, Stefano Ermon, Christopher D. Manning, and Chelsea Finn. **Direct Preference Optimization: Your Language Model is Secretly a Reward Model**, 2024.
- [2] Bram Wallace, Meihua Dang, Rafael Rafailov, Linqi Zhou, Aaron Lou, Senthil Purushwalkam, Stefano Ermon, Caiming Xiong, Shafiq Joty, and Nikhil Naik. **Diffusion Model Alignment Using Direct Preference Optimization**, 2023.

THE DPO INSIGHT: FROM RL TO A SIMPLE LOSS



Reparameterize the reward:

$$r(x, y) = \beta \log \frac{\pi_r(y|x)}{\pi_{ref}(y|x)} + \beta \log Z(x)$$

$\beta \log Z(x)$ **cancels out!**
DPO Objective:

KL-constrained RL's solution:

$$\pi_r(y|x) = \frac{1}{Z(x)} \pi_{ref}(y|x) \exp\left(\frac{1}{\beta} r(x, y)\right)$$

$$\mathcal{L}_{DPO}(\pi_\theta; \pi_{ref}) =$$

$$-\mathbb{E}_{(x, y_w, y_l) \sim \mathcal{D}} \left[\log \sigma \left(\beta \log \frac{\pi_\theta(y_w | x)}{\pi_{ref}(y_w | x)} - \beta \log \frac{\pi_\theta(y_l | x)}{\pi_{ref}(y_l | x)} \right) \right]$$

$$\nabla_\theta \mathcal{L}_{DPO}(\pi_\theta; \pi_{ref}) = -\beta \mathbb{E}_{(x, y_w, y_l) \sim \mathcal{D}} \left[\underbrace{\sigma(\hat{r}_\theta(x, y_l) - \hat{r}_\theta(x, y_w))}_{\text{higher weight when reward estimate is wrong}} \left[\underbrace{\nabla_\theta \log \pi(y_w | x)}_{\text{increase likelihood of } y_w} - \underbrace{\nabla_\theta \log \pi(y_l | x)}_{\text{decrease likelihood of } y_l} \right] \right]$$

Simple Stable Efficient

DIFFUSION-DPO

Denoising Diffusion Model

$$\begin{aligned} p_\theta(x_{0:T}) &= \prod_{t=1}^T p_\theta(x_{t-1}|x_t) \\ p_\theta(x_{t-1}|x_t) &= \mathcal{N}(x_{t-1}; \mu_\theta(x_t), \sigma_{t|t-1}^2 \frac{\sigma_{t-1}^2}{\sigma_t^2} I) \\ L_{DM} &= \mathbb{E}_{x_0, \epsilon, t, x_t} [\omega(\lambda_t) \|\epsilon - \epsilon_\theta(x_t, t)\|_2^2] \\ x_t \sim q(x_t|x_0) &= \mathcal{N}(x_t; \alpha_t x_0, \sigma_t^2 I) \end{aligned}$$

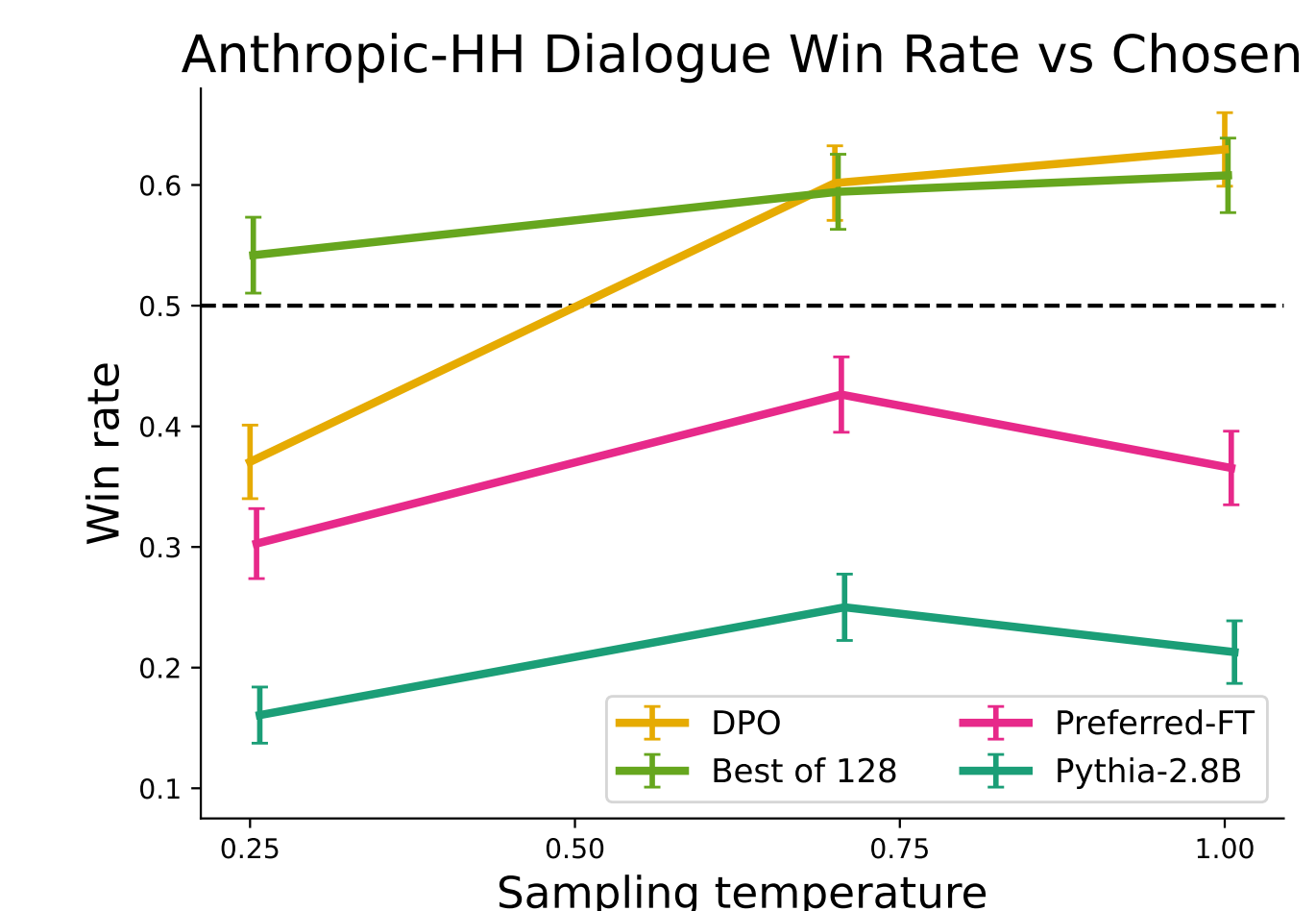
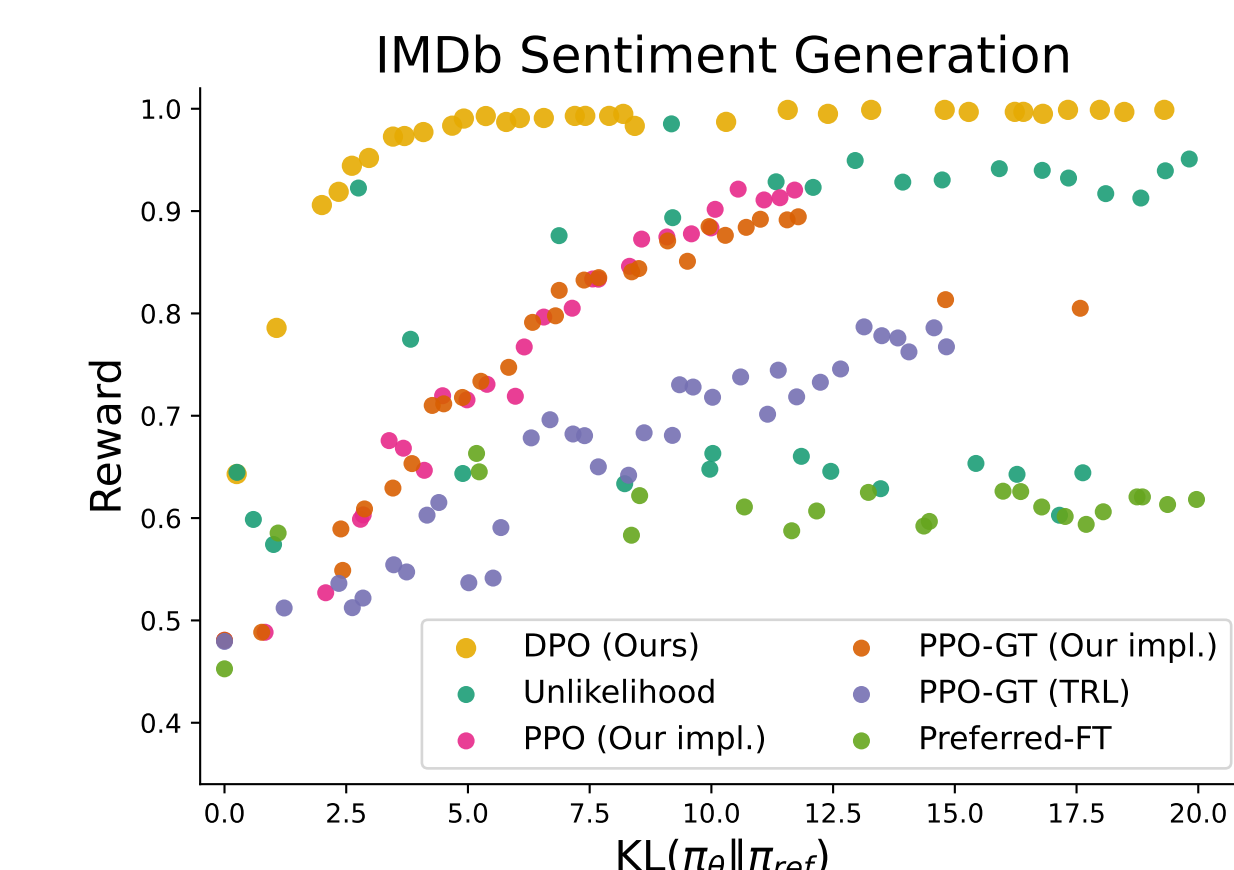
$$L_{DPO}(\theta) = -\mathbb{E}_{c, x_0^w, x_0^l} \left[\log \sigma \left(\beta \log \frac{p_\theta(x_0^w | c)}{p_{ref}(x_0^w | c)} - \beta \log \frac{p_\theta(x_0^l | c)}{p_{ref}(x_0^l | c)} \right) \right]$$

The likelihood of y is intractable $\rightarrow$ ELBO

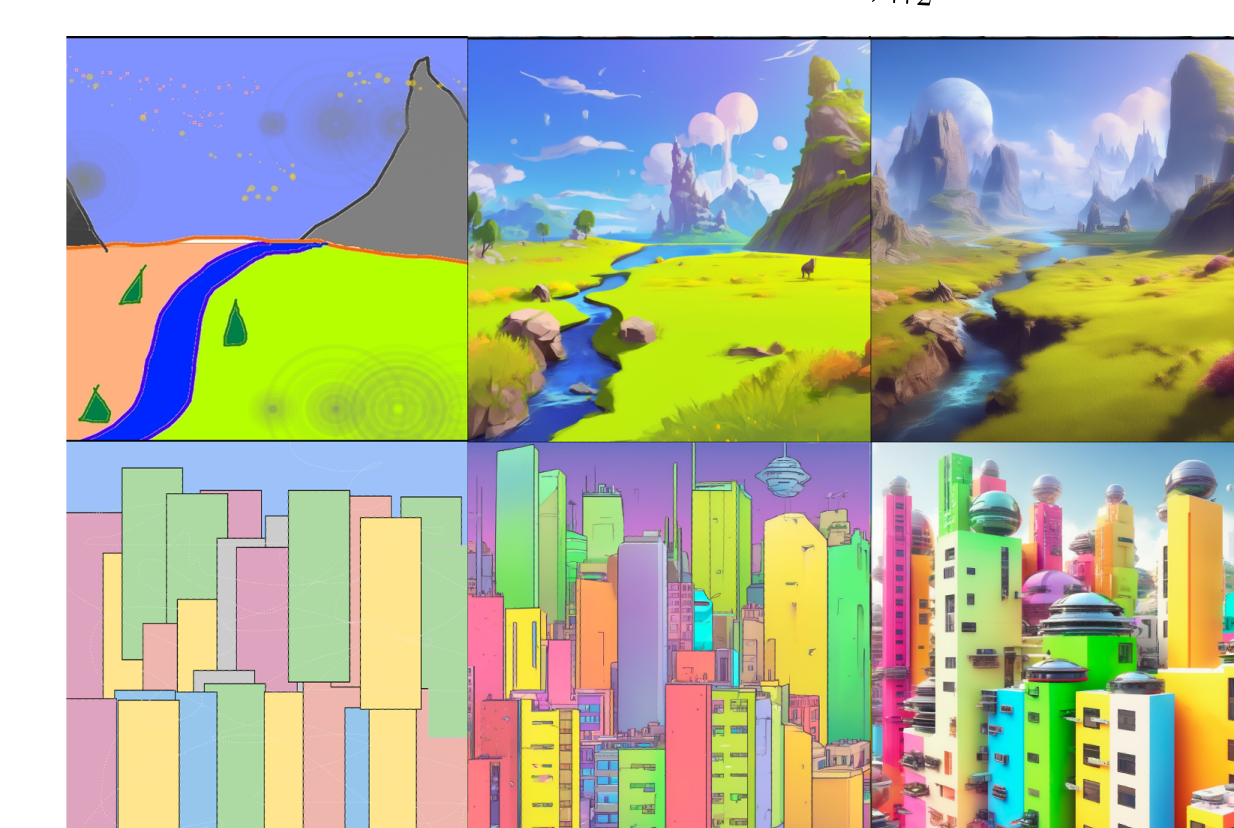
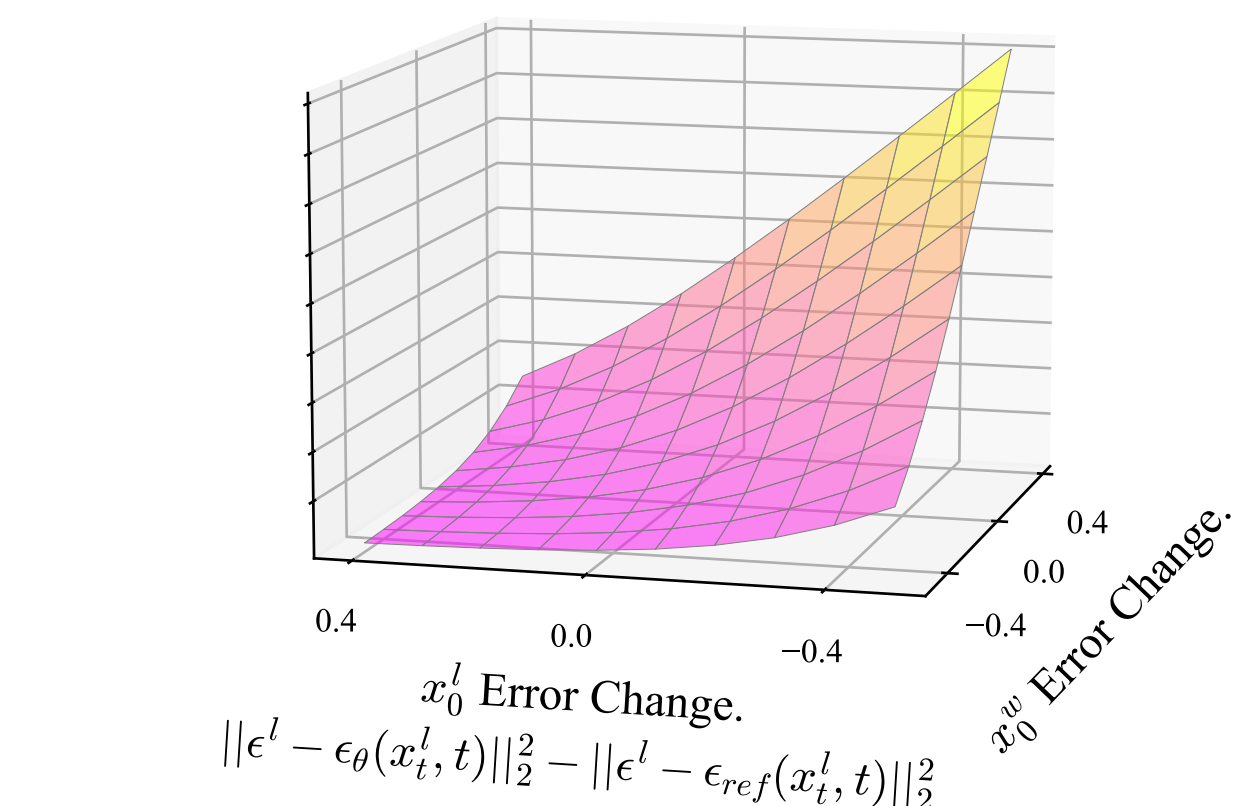
$$\begin{aligned} L_{DPO-Diffusion}(\theta) &\leq -\mathbb{E}_{(x_0^w, x_0^l) \sim \mathcal{D}, t \sim \mathcal{U}(0, T), (x_t^w, x_t^l) \sim q(x_t|x_0)} \\ &\quad \log \sigma \left(-\beta T \omega(\lambda_t) \left(\|\epsilon^w - \epsilon_\theta(x_t^w, t)\|_2^2 - \|\epsilon^w - \epsilon_{ref}(x_t^w, t)\|_2^2 \right. \right. \\ &\quad \left. \left. - \left(\|\epsilon^l - \epsilon_\theta(x_t^l, t)\|_2^2 - \|\epsilon^l - \epsilon_{ref}(x_t^l, t)\|_2^2 \right) \right) \right) \end{aligned}$$



LLM RESULTS



DIFFUSION RESULTS



Original SDXL DPO-SDXL

FUTURE WORKS

Personalization: Fine-tuning models to individual or group-specific preferences (e.g., artistic style).
Learning from AI Feedback: Using a powerful existing model to generate preference labels, enabling massive-scale alignment.
New Modalities: Applying the DPO principle to other generative domains like video, 3D models, or music.

CONTACT INFORMATION

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